

# Forecasting & Evaluation — Easy Guide

Crash course before the presentation · Retail Demand Forecasting (Walmart M5)

## Your topic in one sentence

I explain **how we turn the trained model into a 28-day forecast**, how to read it, and **how we prove the numbers are honest**.

*Not your topic: how the model was trained (that's the ML Engineer). Your topic: how it's **\*used\***, **\*interpreted\*** and **\*verified\***.*

## What we produce

**28 days ahead** (d\_1942–d\_1969, 2016-05-23 → 06-19) for **30,490** product-store combinations = **853,720 numbers**.

## Two ways to forecast, and we use both

**Direct** — one model predicts all 28 days at once. horizon (1–28) is a feature, so it knows which day it's being asked about.

**Recursive** — predict tomorrow, write that prediction back into the history as if it were real, predict the next day. Repeat 28 times.

|                    | Direct         | Recursive  |
|--------------------|----------------|------------|
| Predicts           | all 28 at once | 1 day × 28 |
| Error accumulation | none           | yes        |
| RMSE alone         | 2.1211         | 2.1185     |

## The blend

```
FINAL = clip( 0.60 × DIRECT + 0.40 × RECURSIVE , 0 )
```

**RMSE 2.0929 — better than either member alone.**

Why does that work? Because they're different architectures making **\*different\*** mistakes. Their residual correlation is **0.9488** — high, but not 1, and that gap is the gain.

*The 0.60 weight was chosen on an **inner window ending before** the evaluation window. It was never fitted on the data we report accuracy on. Say this — it's the difference between a real result and a flattering one.*

## The metrics — say these correctly

**RMSE — Root Mean Squared Error** =  $\sqrt{\text{mean}((\text{actual} - \text{predicted})^2)}$  Squaring means **big mistakes hurt much more**. → **2.0929**

**MAE — Mean Absolute Error** =  $\text{mean}(|\text{actual} - \text{predicted}|)$  The typical miss, ignoring direction. → **1.0395**

**WAPE — Weighted Absolute Percentage Error** =  $\text{sum}(|\text{error}|) / \text{sum}(\text{actual})$  We use it instead of MAPE because MAPE divides by the actual, and 68% of our actuals are zero. → **0.7205** (so accuracy =  $1 - 0.7205 = 27.9\%$ )

**Bias** =  $\text{mean}(\text{predicted} - \text{actual})$  → **-0.0224** (essentially neutral)

|                       | RMSE    | MAE            |
|-----------------------|---------|----------------|
| Punishes big errors   | heavily | proportionally |
| Sensitive to outliers | yes     | no             |

RMSE (2.09) > MAE (1.04) tells you a minority of large misses does much of the damage — normal for spiky retail data.

## The "will it sell at all?" metrics

Different question: not \*how many\*, but \*yes/no\*. Rule: actual = sales > 0; predicted = forecast ≥ 0.5 units.

| Metric    | Value         | Meaning                                  |
|-----------|---------------|------------------------------------------|
| Accuracy  | <b>0.6980</b> | how often we were right overall          |
| Precision | <b>0.6321</b> | when we said it'd sell, how often it did |
| Recall    | <b>0.8068</b> | of real sales, how many we caught        |
| F1        | <b>0.7088</b> | one number balancing precision & recall  |
| Base rate | 0.4556        | how often anything sells at all          |

Recall > precision: we catch most real sales but over-flag some quiet days. For retail that's the safer direction.

**Careful — these four get transposed constantly.** Some earlier briefs listed accuracy as 0.8068 and precision as 0.7088 — those are actually the **recall** and **F1**. Use the table above; it comes from GET /api/v1/accuracy/occurrence.

## THE table to memorise — accuracy by level

| Level                         | Groups        | Accuracy     |
|-------------------------------|---------------|--------------|
| Whole chain                   | 1             | <b>97.2%</b> |
| State                         | 3             | 96.0%        |
| Category                      | 3             | 96.2%        |
| Department                    | 7             | 94.8%        |
| Store                         | 10            | <b>94.7%</b> |
| Item (all stores)             | 3,049         | 70.8%        |
| Store-item (what we forecast) | <b>30,490</b> | <b>27.9%</b> |

**Why does it climb?** Errors are largely independent, so they **cancel** when you add them up. One store over-predicts by 2, another under-predicts by 2 — the chain total is exact. At the bottom there's nothing to cancel against, and 54% of individual cells are zero.

*If a judge says "27.9% is low" — this table is your answer. It's not a weak model, it's what per-product-per-day accuracy honestly looks like. Always quote the level that matches the decision.*

## Error grows across the horizon

| Day | RMSE          |
|-----|---------------|
| 1   | 1.8106        |
| 28  | <b>2.3807</b> |

Day 28 is ~31% worse than day 1. We publish that instead of averaging it away.

## The shaded band on the chart

It is **NOT** a confidence interval. Our model gives a single point forecast, not a probability distribution.

It is observed past error: residuals from 8 backtest windows, normalised by `sqrt(forecast)`, grouped by demand regime and horizon day.

**Measured coverage: 90.0%** — it contained the real value 90% of the time.

Every API response carries a `band_basis` field saying this in words.

## The rule we never break

*We never quote accuracy for the delivered forecast window.*

`d_1942–d_1969` has **no ground truth** — nobody knows what actually sold. Every number we report comes from **held-out validation windows**.

## How we prove it reproduces

`POST /api/v1/inference/verify` reloads the frozen models, rebuilds features, re-runs the blend, compares to the shipped file.

`max_abs_diff = 0.000e+00` across all 853,720 predictions. Identical, not "close".

Also: 4 windows validated (2.0929 / 2.1547 / 2.1074 / 2.1496), leakage suite 46/46, and an audit retrained all four windows from scratch and reproduced them to four decimals.

## Baselines — always quote one

|                               | RMSE   |
|-------------------------------|--------|
| Our blend                     | 2.0929 |
| Naive "repeat 28-day average" | 2.2430 |
| Predict zero everywhere       | 3.9161 |

## How to explain it to a judge

"My part is what happens after the model exists — producing the forecast and proving it's trustworthy.

We forecast 28 days for 30,490 product-store combinations — 853,720 numbers. We do it two ways: a direct model predicting all 28 days at once, and a recursive one predicting a day and feeding it back. Alone they're 2.1211 and 2.1185 RMSE; blended 60/40 we get 2.0929, better than either, because they make different mistakes.

RMSE is 2.0929 — it squares errors, so big misses count more. MAE is 1.0395, so the typical miss is about one unit per product per day. Bias is  $-0.0224$ , so we're not systematically off in either direction.

The table I'd show you is accuracy by level: 27.9% at individual product-store-day, 94.7% at store, 97.2% chain-wide. That's not hiding a bad number — errors cancel when you aggregate, and 54% of individual cells are zero. We show whichever level matches your decision.

We never quote accuracy for the delivered forecast, because that window has no ground truth. And we can prove the model still produces exactly what we shipped — re-running it gives a maximum absolute difference of exactly zero."

## Quick answers

\*Is 27.9% good?\* At that level, yes — it's 97.2% chain-wide. Errors cancel on aggregation and 54% of cells are zero.

\*Why WAPE not MAPE?\* MAPE divides by the actual; 68% of ours are zero.

\*What's the shaded band?\* Observed past error, 90.0% measured coverage — not a confidence interval.

\*How accurate will next month be?\* Unknowable — no ground truth for that window. Our honest estimate is the held-out result, RMSE 2.0929.

\*Why blend?\* Residual correlation 0.9488 — different mistakes partly cancel.

\*How was 0.60 chosen?\* On an inner window that ends before the evaluation window.

\*Does error grow with horizon?\* Yes — 1.8106 on day 1, 2.3807 on day 28.

## 30-second revision

28 days · 30,490 series · 853,720 predictions · direct (2.1211) + recursive (2.1185) blended 0.60/0.40 → **RMSE 2.0929**, MAE **1.0395**, WAPE **0.7205**, bias **-0.0224** · occurrence: acc **0.6980**, prec **0.6321**, recall **0.8068**, F1 **0.7088** · accuracy 27.9% store-item → 94.7% store → 97.2% chain · band = observed error, 90.0% coverage · re-inference max\_abs\_diff = 0.000e+00 · never quote accuracy on the delivered window.